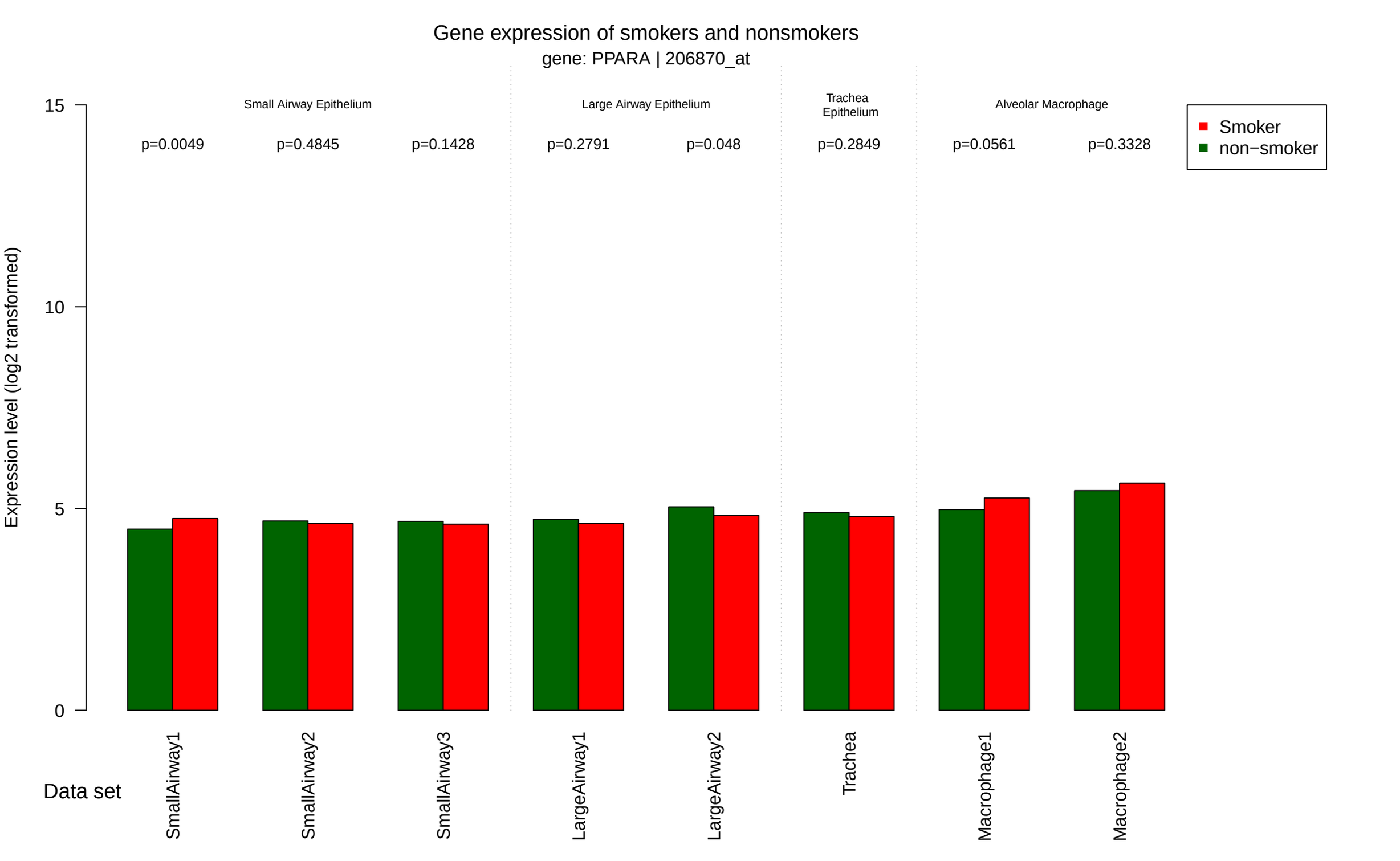
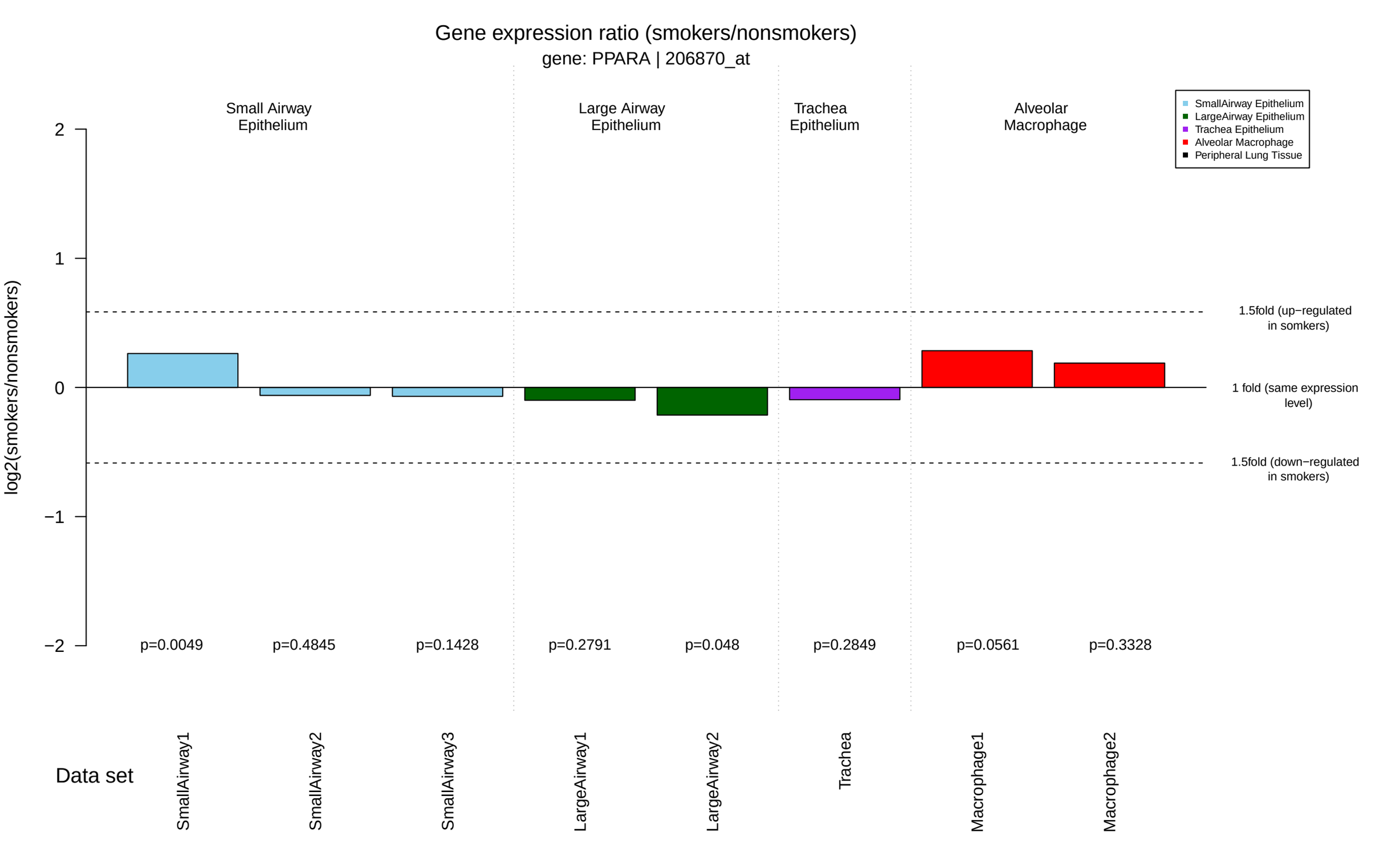
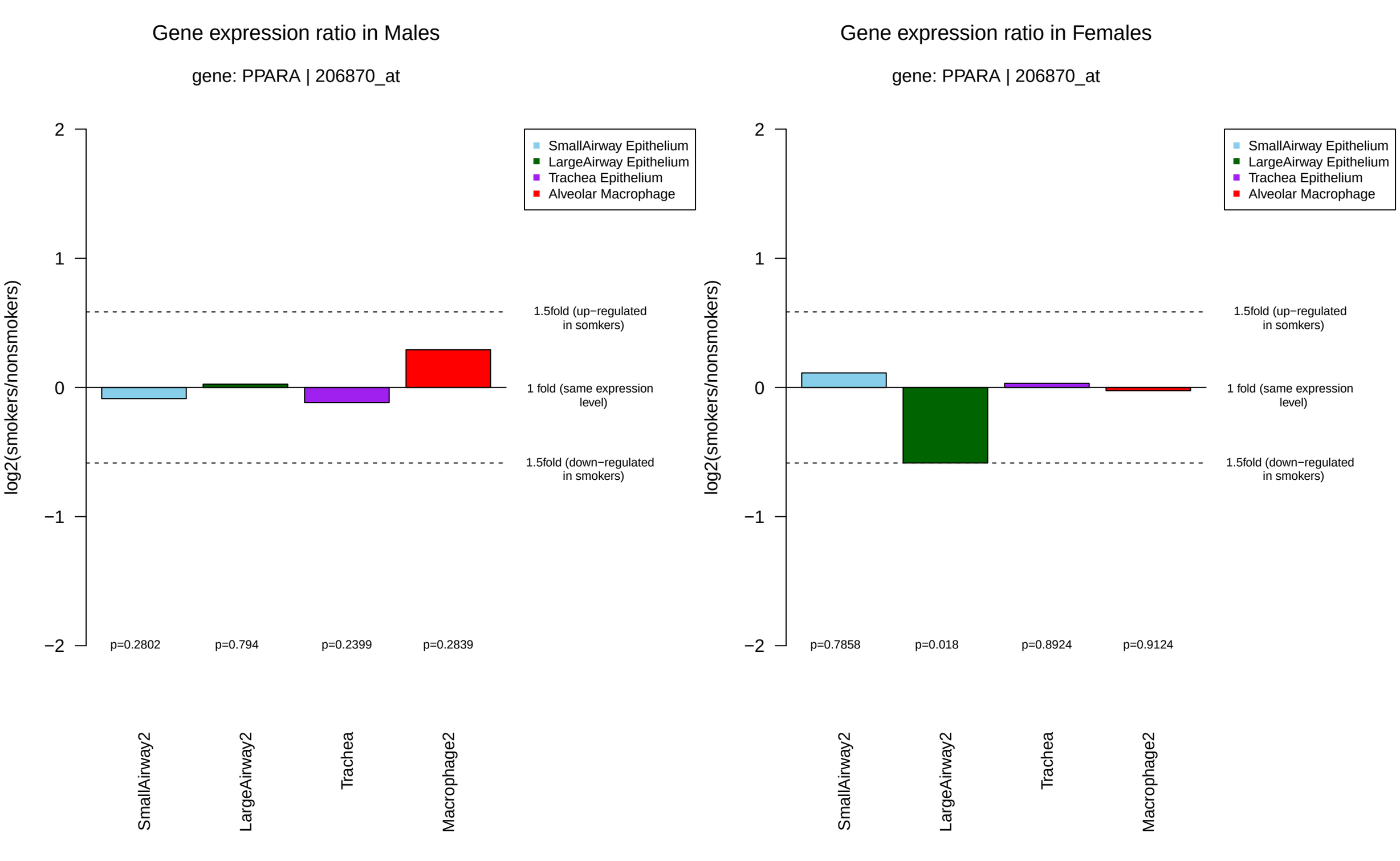
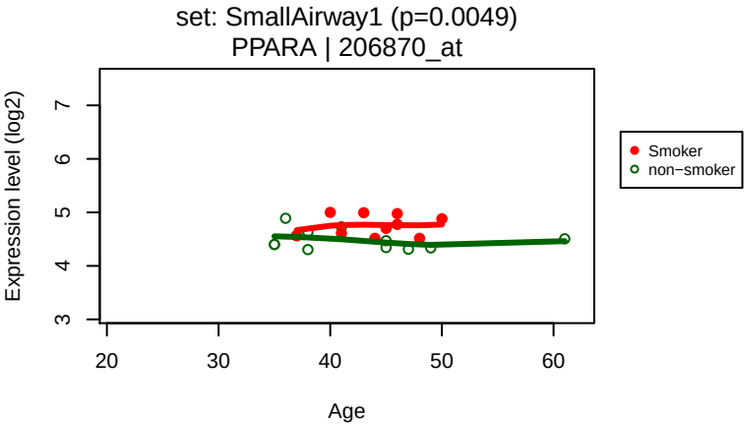
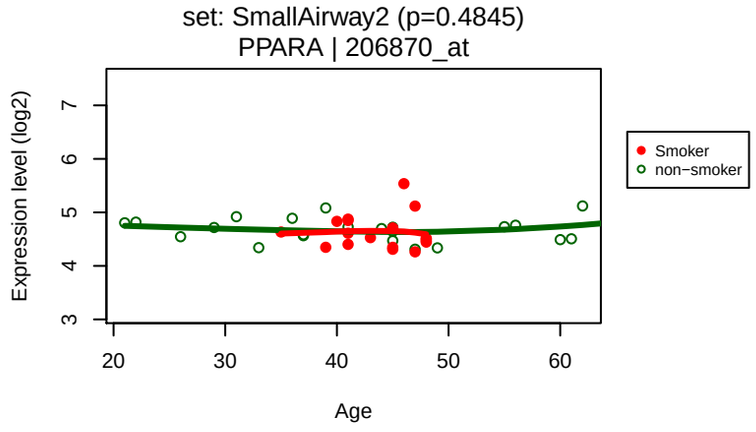
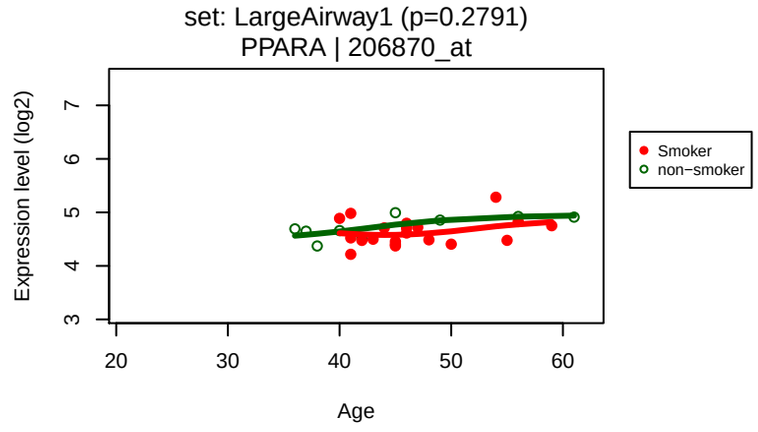

Gene expression of smokers and nonsmokers
gene: PPARA | 206870_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0049
p=0.4845
p=0.1428
p=0.2791
p=0.048
p=0.2849
p=0.0561
p=0.3328
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

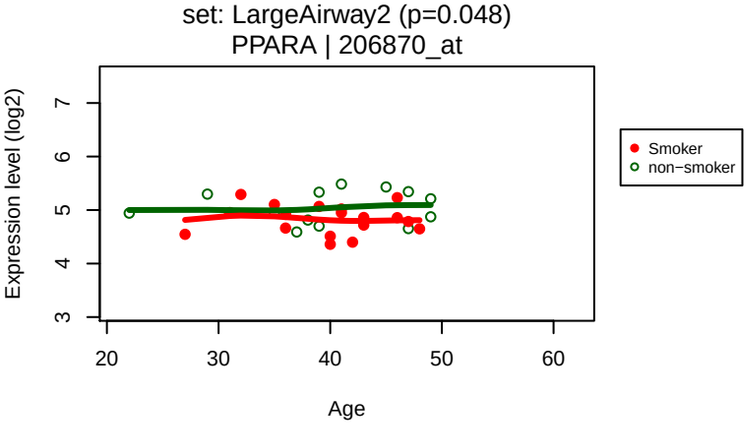

Gene expression ratio (smokers/nonsmokers)
gene: PPARA | 206870_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0049
p=0.4845
p=0.1428
p=0.2791
p=0.048
p=0.2849
p=0.0561
p=0.3328
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

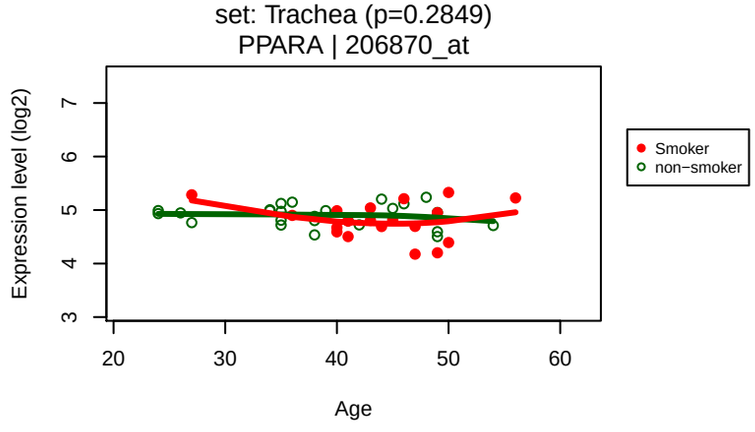

Gene expression ratio in Males
gene: PPARA | 206870_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2802
p=0.794
p=0.2399
p=0.2839
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PPARA | 206870_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7858
p=0.018
p=0.8924
p=0.9124
SmallAirway2
LargeAirway2
Trachea
Macrophage2

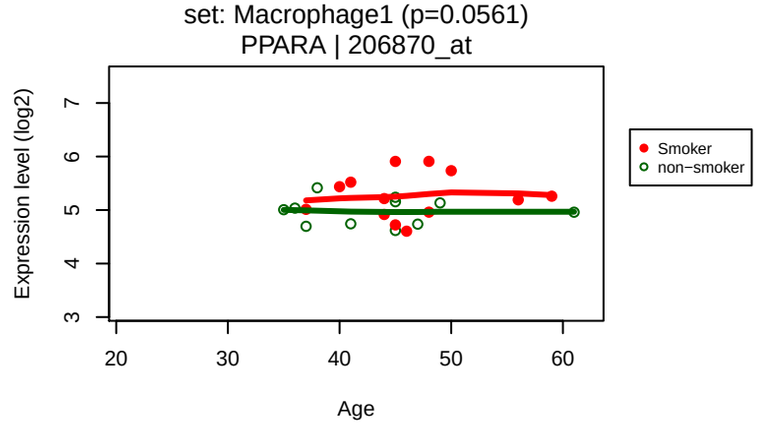

set: SmallAirway1 (p=0.0049)
PPARA | 206870_at
Expression level (log2)
Age
Smoker
non−smoker

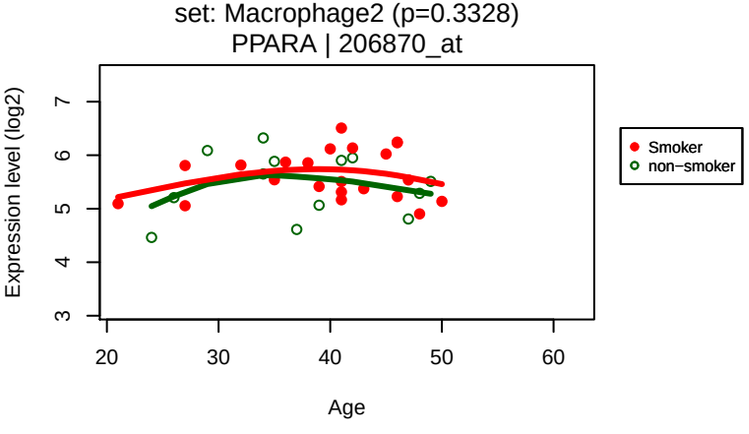

set: SmallAirway2 (p=0.4845)
PPARA | 206870_at
Expression level (log2)
Age
Smoker
non−smoker

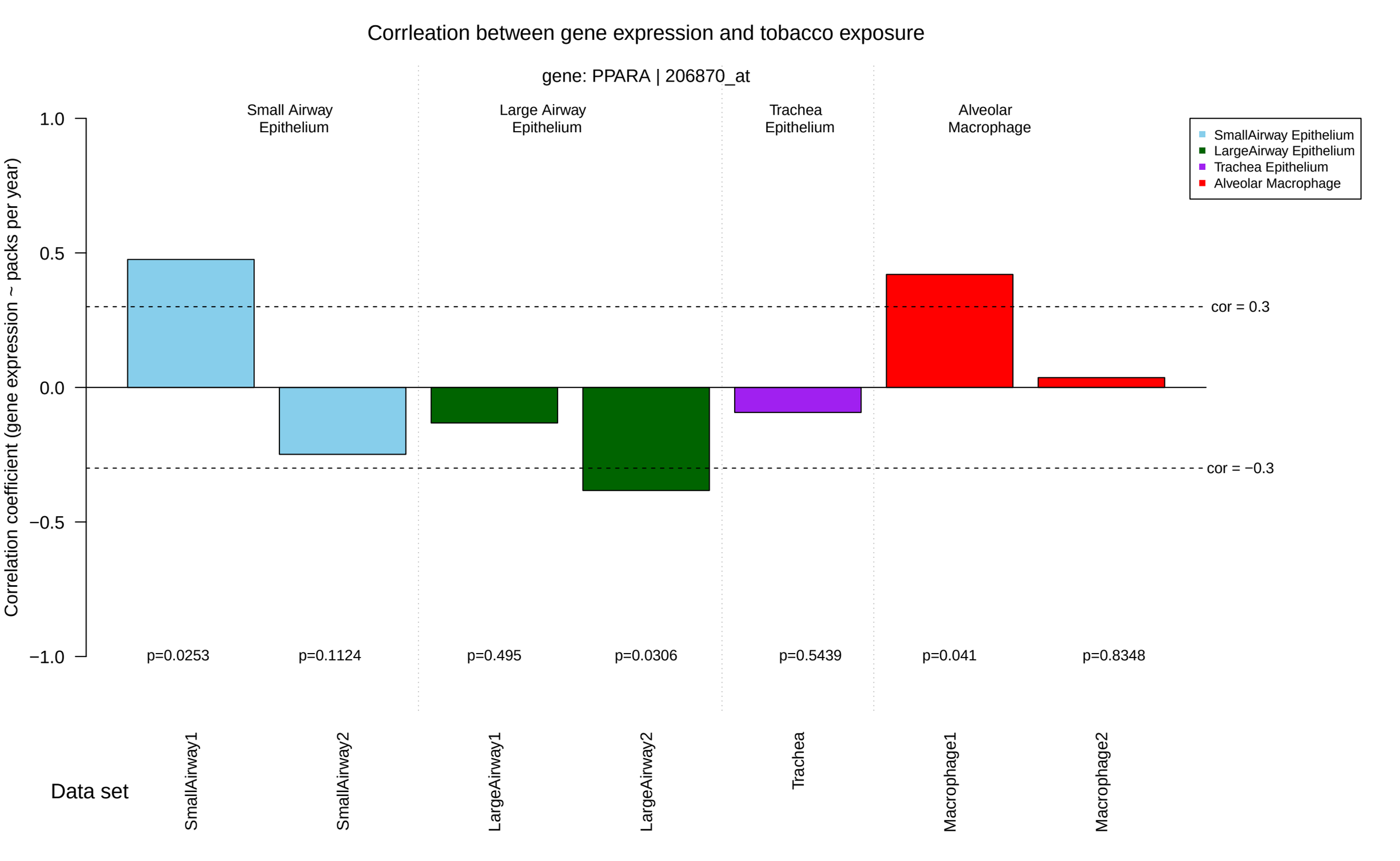

set: LargeAirway1 (p=0.2791)
PPARA | 206870_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.048)
PPARA | 206870_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2849)
PPARA | 206870_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0561)
PPARA | 206870_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3328)
PPARA | 206870_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PPARA | 206870_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0253
p=0.1124
p=0.495
p=0.0306
p=0.5439
p=0.041
p=0.8348
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2